

# Figures that preserve evidence

Project: Reproducible AI Research Study

## 1. Your mission

### Your learning target

I can explain the mechanism and terminology of figures that preserve evidence.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of figures that preserve evidence

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Negative results and claim revision. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

#### My starting explanation

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### This week's end goal

Explain figures that preserve evidence using the records and boundaries of this fictional investigation.

## 2. Notice the evidence

Figures should preserve denominators, scales and uncertainty rather than exaggerate the preferred comparison. A plot is a representation of evidence, not extra evidence.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

**List the given facts and one fact that is not supplied.**

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## 3. See how it works

Label axes, units, cases and scope. For simple comparative bars use a zero baseline or explicitly justify another plot. Show condition breakdowns and unassessed counts where relevant.

### Figures that preserve evidence — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** Rates A 80%, B 82% are plotted on bars starting 79%, making B look several times larger.

**Check or calculation:** Label axes, units, cases and scope. For simple comparative bars use a zero baseline or explicitly justify another plot. Show condition breakdowns and unassessed counts where relevant.

**Conclusion and limit:** Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled difference plot.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### Compare a familiar case with a changed case

Rates A 80%, B 82% are plotted on bars starting 79%, making B look several times larger. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

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### Change one thing in the pictured case

Change: Start the axis at 5 without making the truncation obvious.

Prediction: The bars exaggerate the visual contrast and require correction or clear disclosure.

Reason: Visible heights no longer correspond to the full count scale.

## 5. Words that unlock the idea

### Word · Meaning

**Word:** Axis scale

**Meaning:** The mapping between values and plotted positions

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**Word:** Denominator label

**Meaning:** The sample size underlying a proportion

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**Word:** Difference plot

**Meaning:** A display of a defined comparison magnitude

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**Use two terms to explain a step in this week's example.**

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## 6. Follow a worked example

Rates A 80%, B 82% are plotted on bars starting 79%, making B look several times larger.

Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled difference plot.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Figures that preserve evidence

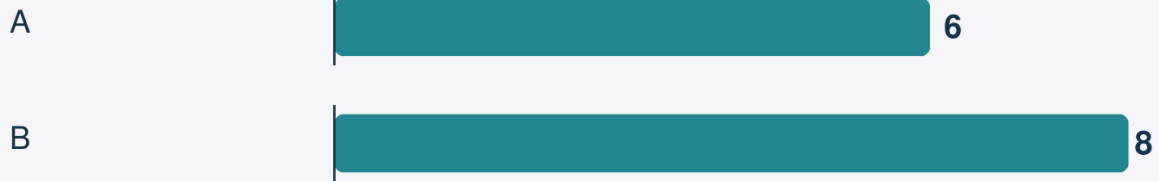
Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.

### The facts we will use

A has 6/10 correct; B has 8/10. The count axis should start at zero and identify the ten-case denominator.

## Zero-based correct counts

correct cases out of 10



Bar length shows the supplied value; the marked baseline is zero.

## Figures that preserve evidence

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Draw honest bars

Use heights 6 and 8 on a zero-based correct-count axis.



### 2. Label the scale

State counts out of ten, not an unspecified quality score.



### 3. Keep context nearby

Note the fixed tasks and absence of a significance test.

### CHANGE ONE THING

Start the axis at 5 without making the truncation obvious.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Draw honest bars

**What happens and why:** Use heights 6 and 8 on a zero-based correct-count axis.

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**Step:** 2. Label the scale

**What happens and why:** State counts out of ten, not an unspecified quality score.

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### Step: 3. Keep context nearby

**What happens and why:** Note the fixed tasks and absence of a significance test.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Figures make evidence easier to inspect when labels and scales preserve meaning. A dramatic-looking visual cannot supply missing uncertainty or representativeness.

### What this example does not establish

A well-labelled chart still cannot establish a causal or population claim from ten cases.

Read the labelled picture: Draw count bars for 6/10 and 8/10 using the stated axis. Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

Change one thing: Start the axis at 5 without making the truncation obvious. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

Visual choices must reflect the intended metric and cannot imply causal certainty from association.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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### Locate the earliest unsupported step

A report omits this qualification: Visual choices must reflect the intended metric and cannot imply causal certainty from association. Explain how omitting it could change a reader's interpretation, and write an

accurate replacement.

**State one repair and the checks you would rerun.**

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## 8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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**Explain your prediction to a partner before comparing results.**

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### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Figures that preserve evidence”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

**Plan · Do · Check**

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for Figures that preserve evidence. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

## Explain the picture

Draw count bars for  $6/10$  and  $8/10$  using the stated axis. Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### Predict, investigate and preserve the evidence

Rates A 80%, B 82% are plotted on bars starting 79%, making B look several times larger. Show the intermediate steps and give the conclusion with its scope.

### Record · Your evidence

**Record:** Given input and fixed rule

**Your evidence:** \_\_\_\_\_

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**Record:** Prediction before checking

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Actual result, or “not run”

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** First mismatch and one explanation

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Boundary or missing input and expected response

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

### Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

### Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: Visual choices must reflect the intended metric and cannot imply causal certainty from association. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

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## 11. Connect your project

Add a labelled figures that preserve evidence evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of figures that preserve evidence. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### Project evidence · Record

**Project evidence:** Component and version

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Inputs permitted at prediction time

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Trace and comparison with earlier work

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Failure, limit and human review

**Record:** \_\_\_\_\_

\_\_\_\_\_

### Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

## 12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside

any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

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### 13. Study a complete case

#### Complete evidence record

#### Figures that preserve evidence — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** Rates A 80%, B 82% are plotted on bars starting 79%, making B look several times larger.

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**Conclusion and limit:** Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled difference plot.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for

qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

### **My complete trace of the worked case**

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## **14. Try a new case independently**

### **Independent case: Apply a new case independently**

CountsA8/10, B9/15 are plotted as8 versus9 without denominators. Correct.

### **My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### **Attempt this without hints**

CountsA8/10, B9/15 are plotted as8 versus9 without denominators. Correct.

### **My independent procedure, result and limitation**

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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

**Use feedback to improve**

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

**15. Your lesson route**

Start with this question: Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.

**By the end, I can · How I will show it**

**By the end, I can:** Explain the mechanism and terminology of figures that preserve evidence

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Apply the stated procedure to a new case with a traceable result

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Identify an assumption or limitation and design a revealing follow-up

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of figures that preserve evidence

**My deciding step, evidence and one remaining question**

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.